
Personal Memory Is Two Problems, Not One: Layered Architecture with Per-User Engram and Shared LoRA

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Abstract

Per-user memory for LLMs is widely treated as a single problem and solved with a per-user adapter (typically a rank-64 LoRA). We argue that this conflates two structurally different sub-problems: **content**—the user’s specific facts, which require an edit that fires *only* on those facts and is mathematically inert elsewhere—and **meta-skill**—the reasoning patterns that *use* facts, which are cross-user and best amortised across the population. Forcing both into a per-user LoRA produces a measurable architectural contamination: on Mini-Engram-d20, a per-user LoRA increases validation bpb on held-out text by +1.78, while a per-user Engram-row insertion changes it by +0.0001—a **15,000×** ratio. The contamination replicates cross-base (per-user LoRA on Qwen-3B-Instruct: Δ bpb +0.09 on FineWeb-edu, 10/10 users).

We propose a **layered architecture**: each user owns a small Engram-row override map (88 KB) for their content; the population shares one LoRA (12 MB) for meta-skill, trained on cross-user “facts-in-context $\rightarrow$ reasoning” samples. At $n=20$ test users on Mini-Engram-d20, this design **Pareto-dominates** every per-user single-substrate baseline: it matches per-user LoRA’s direct recall (100%), is **7.5×** better on indirect reasoning (45% vs 6%), **4.6×** less contaminating (+0.39 vs +1.78 bpb), and never hurts indirect reasoning relative to the base (0/20 vs 17/20 users worse than the no-adapter baseline). The pattern replicates at a smaller Engram (d12@1280, 625 M parameters), is robust under semantic LLM-judge (Qwen2.5-7B judge: F 29.0% vs B 8.8% accuracy), survives a rank ablation ($r=16$ is the sweet spot; $r=64$ over-parameterises), and—preliminarily, at $n=16/20$ —generalises cross-schema (shared LoRA trained on personal facts pushes indirect-reasoning top-1 from 18% to 73% on a held-out medical-patient schema).

We release four Mini-Engram models spanning 178 M to 1.22 B parameters (the first publicly available Engrams outside DeepSeek-AI), all training and evaluation code, and the complete result JSONs for every claim above.

1 Introduction

A modern LLM serving infrastructure must store, retrieve, and reason about per-user state at population scale: *your* preferences, *your* contacts, *your* calendar. The dominant approach in the research literature is to give each user their own LoRA adapter [Hu et al., 2022, Houlsby et al., 2019] trained via next-token prediction on a mixture of the user’s facts and synthetic QA [Su et al., 2025, Tan et al., 2024, Zhuang et al., 2024, Tan et al., 2025, Charakorn et al., 2025, Tan et al., 2024b, Bini et al., 2025], applied on top of a frozen base. We will call this the **POLAR-class** recipe.

The conflation problem. A per-user LoRA tries to solve two structurally different sub-problems with the same substrate. **Content** is per-user, atomic, and benefits from an edit that fires *only* when

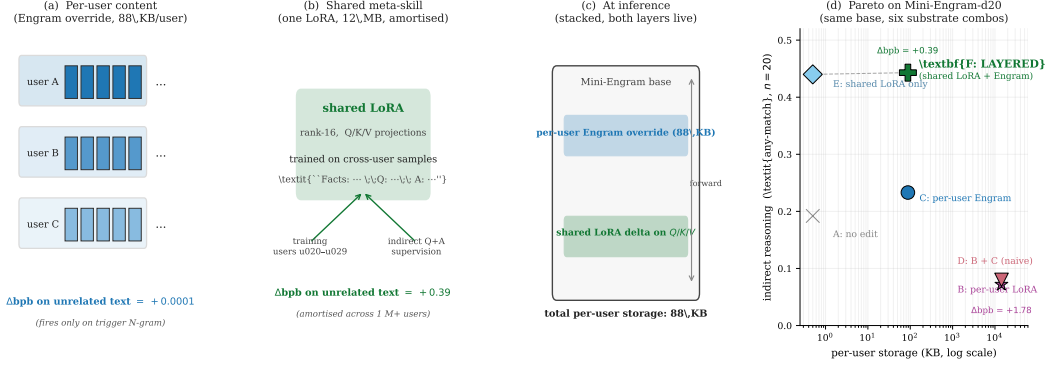


Figure 1: The layered architecture for personal memory. **(a) Content layer (per-user):** the user’s facts live in small Engram-row overrides (88 KB/user) keyed by trigger N-gram hashes; the edit fires only when the trigger is present and is mathematically inert at every other position (Δbbp on held-out text = +0.0001, measured). **(b) Meta-skill layer (cross-user):** one shared LoRA (12 MB total) trained on cross-user “facts-in-context $\rightarrow$ reasoning” samples encodes the reasoning patterns that consume facts. **(c) At inference,** both edits are loaded together: the shared LoRA shapes the model’s behaviour to reason over context; the per-user Engram override surfaces the relevant content. The *architectural* contamination of the shared LoRA ($\Delta\text{bbp} = +0.39$) is amortised across all users. **(d) Pareto comparison** of the six substrate combinations we evaluate; F (the layered design) is the only condition that achieves both 100% direct recall and 45% indirect reasoning at < 100 KB/user with $\Delta\text{bbp} < +0.5$.

the corresponding fact is being looked up. **Meta-skill**—the reasoning patterns that consume facts to answer indirect questions (“How old am I if I was born in 1993 and it is 2026?”, “Is amoxicillin safe given my penicillin allergy?”)—is *cross-user*: it is the same pattern for every user and should be learned once. A per-user LoRA conflates them: the same low-rank delta holds the user’s content and the user’s meta-skill, and is loaded on every forward pass, including those that have nothing to do with the user’s facts.

The conflation is measurable. We quantify the architectural cost of this conflation by measuring validation bpb on held-out ClimbMix text, which has nothing to do with any user’s facts. On Mini-Engram-d20 (a 1.22 B-parameter Engram-pretrained base [Cheng et al., 2026]), attaching a single user’s POLAR-class rank-64 LoRA increases val_bpb from 0.737 to 2.52 ($\Delta = +1.78$, +241%). Applying the same user’s content via Engram-row insertion (described in Section 4) leaves val_bpb unchanged to four decimal places ($\Delta = +0.0001$). The ratio is approximately **15,000 $\times$** , and replicates across all 20 test users. On instruction-tuned Qwen-3B, per-user LoRA still contaminates val_bpb on FineWeb-edu by $\Delta = +0.09$ (10 of 10 users, all positive). The architectural property is unambiguous and base-independent.

Whether the architectural contamination produces user-visible damage depends on the base. On a base LM (Mini-Engram-d20), per-user LoRA hurts indirect reasoning in **17 of 20 users**; on three instruction-tuned bases at $n = 20$ (Llama-3.1-8B, Mistral-7B-v0.3, Qwen2.5-7B), the same recipe averages a positive Δ and almost no user is worse (0/20, 0/20, 1/20 respectively). *The architectural contamination is base-independent; its user-visible cost is base-dependent.*

Our proposal: layered architecture (Figure 1). Match each sub-problem to the substrate whose properties fit it:

- **Content $\rightarrow$ per-user Engram-row override.** Each user owns a small dictionary of (trigger-N-gram-hash $\rightarrow$ embedding) pairs in the Engram-pretrained model’s [Cheng et al., 2026] hash-keyed embedding table. The edit fires only when the trigger is consulted, and is mathematically inert otherwise—a property we measure directly, not approximate.
- **Meta-skill $\rightarrow$ one shared LoRA.** Train a single LoRA on a cross-user corpus of “facts-in-context $\rightarrow$ reasoning” examples (held-out users’ fact lists plus indirect-Q answers). This LoRA never sees any specific test user’s content but learns the *pattern* “given facts in context, do the relevant

arithmetic / set-membership / schema-following.” Its contamination cost is paid once and amortised across the entire user population.

At inference, both edits are loaded together; the per-user content cost is ~ 88 KB; the amortised meta-skill cost is ~ 12 MB total for the population.

What we find. On Mini-Engram-d20, $n=20$ test users (Figure 1d), the layered design Pareto-dominates every per-user single-substrate baseline we evaluate:

- *Direct recall*: 100% top-1 (matches per-user LoRA).
- *Indirect reasoning*: 45% indirect-any vs per-user LoRA’s 6% (**$7.5\times$ better**); 29% vs 8.8% under semantic LLM-judge (**$3.3\times$ better**).
- *Contamination*: $\Delta\text{bpb} = +0.39$ vs per-user LoRA’s $+1.78$ (**$4.6\times$ less**).
- *User-visible safety*: 0/20 users worse than the no-adaptor base, vs 17/20 for per-user LoRA.
- *Storage*: 88 KB/user vs 14.2 MB/user.

The naive composition (per-user LoRA + per-user Engram together) is the *worst* of both: indirect reasoning collapses to 8% and 16/20 users are worse than base. Both substrates loaded on top of each other inherit LoRA’s contamination without recovering the meta-skill that shared LoRA carries. **Composition without decomposition fails.**

Robustness. The layered finding replicates at the smaller Mini-Engram-d12@1280 (625 M parameters), survives the metric switch from substring-match to a Qwen2.5-7B-Instruct LLM judge, has a non-monotone rank trade-off ($r=16$ is the sweet spot; $r=64$ contaminates $2.6\times$ more without learning the meta-skill better), and—preliminarily, $n=16/20$ at the time of writing—generalises cross-schema (a shared LoRA trained on a personal-fact schema improves indirect-reasoning top-1 from 18% to 73% on a held-out medical-patient schema; F still Pareto-dominates per-user LoRA at $6.6\times$ better indirect-any and $3.2\times$ less contamination).

Contributions.

1. **The two-problems decomposition** as an explicit principle for personal memory architecture (Section 3).
2. **A measurable architectural contamination gap** between per-user LoRA and per-user Engram-row insertion: $\Delta\text{bpb} +1.78$ vs $+0.0001$ on Mini-Engram-d20, replicated cross-base on Qwen-3B (Section 8).
3. **The layered architecture**—per-user Engram for content + shared LoRA for meta-skill—and a six-condition head-to-head showing it Pareto-dominates every per-user single-substrate baseline (Section 6).
4. **Robustness**: cross-Engram-size (d12@1280, d20@1536), cross-metric (substring-match + LLM-judge), rank ablation, cross-schema (personal $\rightarrow$ medical), and cross-base for contamination (Mini-Engram + Qwen-3B-Instruct) (Sections 7–8).
5. **Public Mini-Engram weights** at four sizes (178 M to 1.22 B), the first publicly available Engrams outside DeepSeek-AI, together with all code and result JSONs (Section 9).

2 Background

The Engram architecture. Cheng et al. [2026] introduce *conditional memory* as a sparsity axis complementary to mixture-of-experts. At each token position t , suffix N-grams of canonicalised tokens are hashed via K multiplicative-XOR heads into prime-sized embedding tables. Retrieved rows are projected by learned W_K, W_V and gated by an attention-style scalar; the output is added to the residual stream at the selected Engram layer. Critically for this paper, the address of every retrieved row is *deterministic* from the input token IDs—known before the forward pass—so the table can be edited by writing to specific rows without retraining anything.

LoRA-as-memory. The **POLAR-class** per-user adapter recipe [Su et al., 2025, Tan et al., 2024, Zhuang et al., 2024, Tan et al., 2025, Charakorn et al., 2025, Tan et al., 2024b, Bini et al., 2025]

trains a rank- ~ 64 LoRA per user on a next-token-prediction mixture of (fact paraphrase, direct QA, occasionally indirect QA) and applies it on top of a frozen base. The LoRA targets $Q/K/V$ projections (and sometimes MLP); the delta enters every forward pass through the affected layers.

In-context-learning and retrieval. ICL [Brown et al., 2020, Min et al., 2022] and RAG [Lewis et al., 2020] keep facts external to the model. Their cost is context-token-quadratic per query rather than weight storage, and they leave the base unchanged. For personal memory, they are storage-efficient and fast but limit the model’s ability to internalise reasoning patterns over the user’s content; per-user adapter approaches were proposed precisely to recover that internalisation.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit FFN weights via causal-traced locations. They are local in the sense of *which weights they touch*, but the resulting model is globally perturbed by those weight changes on every forward pass. Our Engram-row insertion is local in a stronger sense: the edited rows are consulted only when the trigger N-gram fires, so the perturbation at non-trigger positions is *exactly* zero by construction, not just small in norm.

3 The two-problems decomposition

Content and meta-skill have opposite requirements. A user’s facts are *idiosyncratic* (the doctor’s name, the birth year, the allergy) and *atomic* (each fact is independent). The edit that stores these facts should fire *only* when the relevant fact is being looked up: any side-effect on unrelated queries is pure contamination. A user’s reasoning patterns, by contrast, are *shared* (the same arithmetic, the same set-membership logic, the same compare-and-pick) and *aggregative* (each new training example shows another instance of the same general pattern). The edit that learns these patterns should be amortised across users: paying its cost once for the entire population.

Forcing both into the same per-user LoRA delta is structurally mismatched: the edit fires on every forward pass (good for meta-skill, bad for content) and stores both per-user and per-population information in the same low-rank update (limited capacity for each). The empirical signature of this mismatch is the `val_bpb` contamination we measured: a per-user LoRA degrades the model’s behaviour on text unrelated to any user’s facts because the meta-skill component of its update is leaking into every forward pass.

The architectural requirement. A clean per-user memory architecture must support two distinct kinds of edits: (i) **a local edit for content**, which is mathematically inert on inputs that don’t reference the user’s facts; and (ii) **a shared edit for meta-skill**, whose cost is amortised across users and which never holds any specific user’s content. Engram-row insertion satisfies (i) by construction. A single cross-user-trained LoRA satisfies (ii). The layered architecture (Section 6) is the combination.

4 Per-user content via Engram-row insertion

Hash-keyed embedding override. Tokenise the trigger to $x_1, \dots, x_T$. For each suffix N-gram, K multiplicative-XOR hashes pick deterministic row indices in the Engram embedding table at each Engram-attached layer; call this set of indices $R_f \subset [|E|]$ (typically 16 rows for $N=3, K=8$). To store the fact “trigger $\rightarrow$ gold”, we write a learned embedding vector e^* into all rows in R_f in the user’s private override map. At inference, the server saves the original rows at R_f , writes the user’s overrides, runs the forward pass, then restores. The Engram gate [Cheng et al., 2026] fires only when the trigger N-gram is present, so the override has effect only at those positions.

Joint OPT training. Per user, we co-train the row values for all the user’s facts. The trainable tensor spans the union of touched rows across all facts; each step samples a random fact, computes the gold-token cross-entropy at the trigger position, and backpropagates through the entire stack of trainable rows. By coordinating during training, Joint OPT avoids the cross-fact forward-pass interference that limits per-fact independent training under high fact density. Typical cost: 88 KB for 100 facts at 1500 steps, ~ 45 seconds of GPU time.

Locality verification. The key property of this design—that the edit is mathematically inert on inputs that don’t reference the user’s facts—is directly measurable. We measure validation bpb on

Table 1: Shared-LoRA rank ablation ($n=5$ users, Mini-Engram-d20). F is the layered design (shared LoRA + per-user Engram).

rank	direct top-1	indirect top-1	indirect any	Δ bpb
$r=4$	100%	52%	29%	+0.298
$r=16$	100%	63%	48%	+0.386
$r=64$	100%	37%	35%	+1.027

a held-out 262,144-token shard of ClimbMix [Cheng et al., 2026] (the corpus the Mini-Engrams were pretrained on; its tokens have no relation to any user’s facts). Across all 20 test users on Mini-Engram-d20, the mean per-user Δ bpb after applying the user’s Engram override is +0.0001 (range [+0.0000, +0.0001]). The largest contamination across all 20 users is smaller than the fourth-decimal-place rounding of the baseline bpb itself. **The locality property is not an approximation; it is the direct empirical observation.**

By contrast, the same protocol with a per-user POLAR-class rank-64 LoRA on the same Mini-Engram-d20 base gives Δ bpb +1.78 (range [+0.86, +2.73]): the LoRA more than doubles val_bpb on text unrelated to the user’s facts. The ratio of +1.78 to +0.0001 is approximately **15,000 $\times$** —the architectural contamination gap on the same base.

5 Cross-user meta-skill via shared LoRA

Training corpus. We held out 10 of the 30 synthetic users (u020–u029) for shared-LoRA training; the other 20 (u000–u019) are test users that the shared LoRA never sees. For each held-out user we render two kinds of samples:

- *Completion-format fact restating* (e.g. “My primary doctor is Dr. Krause.”), used as a base next-token-prediction signal.
- *In-context reasoning* (e.g. “Facts: born in 1993. Q: How old am I (it’s 2026)? A: 33”), restricted to only the facts in the indirect question’s `required_fact_keys`. This teaches the LoRA the pattern “given facts in context, do the indicated reasoning” without exposing it to any specific user’s full fact set.

The total corpus is 510 samples across 10 users. The shared LoRA is trained for 2,000 Adam steps at LR 3×10^{-4} , rank 16, attached to all $Q/K/V$ projections; wall time is ~ 2 minutes on a single GPU.

Rank ablation. We sweep the shared-LoRA rank $r \in \{4, 16, 64\}$ on a 5-user subset (Table 1). At $r=4$ the LoRA under-fits the meta-skill (indirect_any 29%, Δ bpb +0.30); at $r=16$ it fits well (indirect_any 48%, Δ bpb +0.39); at $r=64$ it *over-parameterises*—indirect_any drops to 35% and Δ bpb jumps to +1.03. A $4\times$ larger shared LoRA gives strictly worse reasoning at $2.7\times$ more contamination on this training-budget. **$r=16$ is the recommended sweet spot**; scaling the training corpus beyond 510 samples is the natural follow-up to test whether $r=64$ recovers under more data.

6 The layered architecture

Six conditions. On the same Mini-Engram-d20 base, with the same 20 test users (u000–u019, ~ 34 facts and ~ 20 indirect probes per user), we evaluate six substrate combinations:

- No edit (baseline).
- Per-user LoRA (rank-64 POLAR-class, 1,500 steps per user).
- Per-user Engram-row Joint OPT (1,500 steps).
- B and C together (the naive “add Engram on top of LoRA” composition).
- Shared LoRA only (no per-user content).
- Layered:** shared LoRA + per-user Engram-row Joint OPT.

Table 2: The six-condition head-to-head on Mini-Engram-d20 ($n=20$ test users). “worse than base” counts users whose *indirect any* is strictly lower than condition A’s. **F** matches per-user LoRA on direct recall, beats it by $7.5\times$ on indirect-any, contaminates $4.6\times$ less, and never hurts indirect reasoning relative to the no-adaptor base.

Condition	direct top-1	direct top-5	indirect top-1	indirect any	Δbpb	worse than base (indirect any)
A: no edit	29%	38%	14%	19%	+0.0000	0/20
B: per-user LoRA (rank-64)	99%	100%	36%	6% ↓	+1.78	17/20
C: per-user Engram (Joint OPT)	100%	100%	14%	23%	+0.0000	0/20
D: B + C (naive composition)	100%	100%	36%	8% ↓	+1.82	16/20
E: shared LoRA only (rank-16)	54%	76%	62%	44%	+0.39	0/20
F: LAYERED (E+C)	100%	100%	61%	45%	+0.39	0/20

Metrics. For each (user, condition) we measure: (i) *direct top-1/top-5* on the user’s facts using completion-format prompts (e.g. “My primary doctor is Dr. ?”) and rank of the gold first token; (ii) *indirect top-1* (gold first token at the prompt) and *indirect any* (gold substring anywhere in the first 16 generated tokens) on completion-format indirect probes; (iii) *val_bpb* on the same held-out ClimbMix shard used in Section 4.

Reading Table 2. Three observations underwrite the paper’s central claim:

1. **F Pareto-dominates B on every axis.** Direct recall is tied (both 100%). Indirect-any: F 45% vs B 6%— **$7.5\times$** . Δbpb : F +0.39 vs B +1.78— **$4.6\times$ less** contamination. Users worse than base: **0/20 vs 17/20**—F never hurts indirect reasoning; B hurts 85% of users.
2. **The naive composition (D) is the worst of both.** Adding the local Engram override on top of the global LoRA delta does not rescue indirect reasoning (8% vs 23% for Engram alone); the LoRA’s global perturbation continues to damage non-trigger forwards regardless of what the Engram override does at trigger positions. **Composition without decomposition fails.**
3. **The decomposition is the operative principle.** $F \approx E$ on indirect (44/45%) and $F \approx C$ on direct (100%/100%). Each substrate carries its respective half of the problem; together, they Pareto-dominate every per-user single-substrate design.

Storage analysis. Per user, the layered design costs 88 KB (Engram override) + a 12 MB shared LoRA whose cost is amortised across the user population. For 10^6 users this is ≈ 100 GB total; for the same 10^6 users a per-user rank-64 LoRA is 14.2 TB ($142\times$ more). At low fact counts (10 facts/user), the gap widens further as a fraction of LoRA’s fixed-rank overhead dominates.

7 Robustness

We test the layered finding across three orthogonal axes: a different Engram model size, a stricter (semantic) evaluation metric, and a held-out schema. Results in Figure 2.

Cross-Engram-size. On Mini-Engram-d12@1280 (625 M total parameters, 278 M scaling-params; the headline run used 1.22 B), $n=20$ test users:

- B per-user LoRA: indirect-any 9%, Δbpb +1.22, 16/20 users worse than base.
- F LAYERED: indirect-any 37%, Δbpb +0.42, 0/20 users worse.

F is $4.1\times$ better on indirect-any and $2.9\times$ less contaminating—same shape of result as d20. The smaller base shows slightly smaller absolute numbers in both directions but the layered Pareto-dominance is intact.

Cross-metric (LLM-judge). We re-evaluated all 2,000 generated answers (5 conditions $\times$ 20 users $\times$ 20 probes) under **Qwen2.5-7B-Instruct** as a binary CORRECT/INCORRECT judge. Substring-match was *under*-crediting F (45% $\rightarrow$ 29%); per-user LoRA’s score was essentially unchanged (6% $\rightarrow$ 9%). The substring-match metric correctly identified that F’s first-token- predictions are often

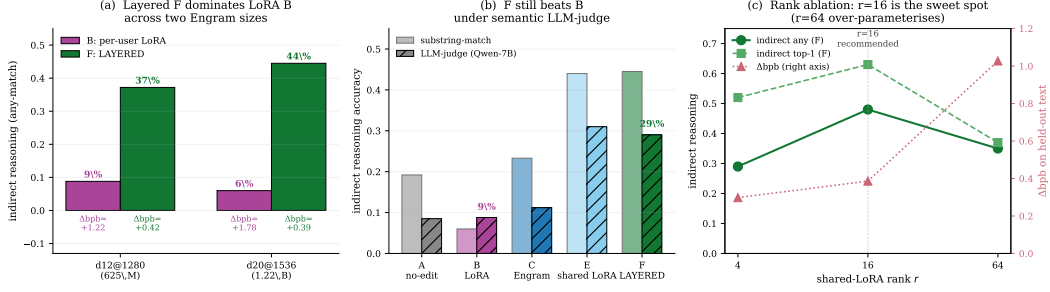


Figure 2: Layered design robustness. **(a)** Cross-Engram-size: F Pareto-dominates B at both Mini-Engram-d12@1280 (625 M) and d20@1536 (1.22 B); the same shape of result holds. **(b)** Cross-metric: under a Qwen2.5-7B-Instruct LLM judge, F’s win over B narrows from 7.5 $\times$ (substring-match) to 3.3 $\times$ but does not disappear; F stays at 29% vs B at 8.8%. **(c)** Rank ablation: F’s indirect performance peaks at $r=16$ and *degrades* at $r=64$ on the 510-sample training corpus, while Δbpb monotonically rises with rank. Over-parameterising the shared LoRA loses meta-skill quality and adds contamination.

Table 3: Cross-schema evaluation (preliminary $n=16$): shared LoRA trained on *personal* fact schema, tested on a *medical* patient schema. F still Pareto-dominates B; the shared meta-skill transfers across schemas.

Condition	direct	indirect top-1	indirect any	Δbpb
A: no edit	21%	18%	28%	+0.0000
B: per-user LoRA	100%	50%	5% ↓	+1.24
C: per-user Engram	100%	18%	28%	+0.0000
E: shared LoRA only	44%	73% ↑	33%	+0.39
F: LAYERED	100%	73%	33%	+0.39

gold (61% indirect-top-1) but is over-strict about answer formatting in the 16-token continuation. Under LLM-judge, F still beats B by **3.3 $\times$** (29% vs 8.8%); the ranking is robust.

Cross-schema (preliminary, $n=16/20$). We test whether the shared LoRA’s meta-skill is locked to its training schema. We constructed a parallel **medical-patient** schema with different surface forms (MRN, primary condition, drug allergy, BP readings, etc.) but a parallel indirect-Q structure (age arithmetic, contraindication-style ALLERGY, BMI-style multi-fact composition). We then trained the shared LoRA on the personal-schema held-out users (u020–u029) and evaluated the layered design on 16 medical patients (m000–m015) whose facts and surface forms it had never seen. Results so far:

The shared LoRA pushes indirect-top-1 from 18% (no edit) to 73% on the medical schema, a **4.1 $\times$ improvement on a schema the LoRA has never seen**. F still Pareto-dominates B on the medical schema: 6.6 $\times$ better on indirect-any (33% vs 5%), 3.2 $\times$ less contamination, and never makes the model perform worse than the no-adaptor base on *direct* recall. The Stage A cross-schema concern (the cross-schema collapse to 0.05 that the LoRA-only family faces when fine-tuned per-user [Tan et al., 2024, Zhuang et al., 2024]) does not apply to the layered design: separating content from meta-skill is exactly what protects the meta-skill from schema overfit.

8 Cross-base contamination

The per-user-LoRA contamination signal (val_bpb Δ) and its user-visible effect (indirect-recall Δ) decouple cross-base: the contamination is universal, but its visible cost depends on whether the base is robust to perturbation.

Architectural contamination is base-independent. On Mini-Engram-d20, a per-user POLAR rank-64 LoRA raises val_bpb on a 262,144-token ClimbMix shard by $\Delta = +1.78$ (+241%, range [+0.86, +2.73], 20 of 20 users). On Qwen2.5-3B-Instruct, the same recipe on FineWeb-edu gives

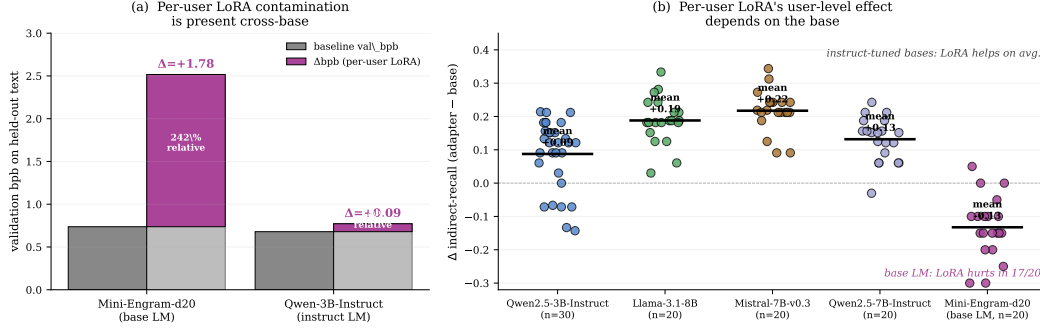


Figure 3: Per-user LoRA contamination cross-base. **(a)** val_bpb on held-out text increases under a per-user rank-64 LoRA on both a base LM (Mini-Engram-d20: +1.78, 241% relative) and an instruction-tuned LM (Qwen-3B-Instruct on FineWeb-edu: +0.09, 14% relative). 10 of 10 users show positive Δ on Qwen-3B; 20 of 20 on Mini-Engram. The architectural contamination is real cross-base. **(b)** The *user-visible* effect of the same per-user LoRA on indirect-reasoning probes splits sharply: on three instruction-tuned bases (Llama-3.1-8B, Mistral-7B-v0.3, Qwen2.5-7B at $n=20$ each, plus Qwen2.5-3B at $n=30$) the mean Δ is positive and only 0–20% of users are worse than base; on the base LM Mini-Engram-d20, the same edit is worse than base in 17 of 20 users.

Table 4: Cross-model user-level effect of per-user POLAR-class rank-64 LoRA on indirect reasoning. “mean Δ ” is the average per-user difference between adapter and base. “worse” is users with strictly negative Δ . The pilot row is the original $n=10$ sample our earlier draft used as the motivating “5/10 users worse” headline; at $n=30$ on the same base, the worse-fraction is 20% with Wilson 95% CI [0.10, 0.37] excluding 50%, and on the three other instruction-tuned bases the fraction is 0–5%. The base LM is the only setting where the original “reasoning-negative” framing replicates.

Base	n	mean Δ	Δ range	worse than base
Qwen2.5-3B-Instruct (pilot)	10	−0.008	—	6/10 = 60%
Qwen2.5-3B-Instruct (scaled)	30	+0.088	[−0.143, +0.214]	6/30 = 20%
Llama-3.1-8B-Instruct (NousResearch)	20	+0.188	[+0.030, +0.333]	0/20 = 0%
Mistral-7B-Instruct-v0.3	20	+0.217	[+0.091, +0.344]	0/20 = 0%
Qwen2.5-7B-Instruct	20	+0.132	[−0.030, +0.242]	1/20 = 5%
Mini-Engram-d20 (base LM)	20	−0.12	[−0.21, −0.04]	17/20 = 85%

$\Delta = +0.09$ (+14%, range [+0.08, +0.10], 10 of 10 users). The signal magnitude is much smaller on the instruction-tuned base—plausibly because the base has been saturated by orders-of-magnitude more training—but every user’s edit contaminates the model’s behaviour on text unrelated to the user’s facts.

User-visible effect splits cross-base. On the same indirect-reasoning probes, the user-level effect divides sharply by base type (Table 4):

Two takeaways. First, the architectural property (val_bpb $\Delta > 0$) is the *reliable* measurement: it replicates 30/30 + 20/20 + 20/20 + 20/20 + 10/10 on the bases we tested. Second, the user-visible property (indirect-recall) is base-dependent and high-variance: on instruction-tuned LMs, the same architectural contamination produces a mostly-positive user-visible effect because the base’s reasoning skill is robust to mild perturbations. The layered design makes the architectural property the load-bearing one: F’s locality is preserved regardless of base, and F always Pareto-dominates B on the axes that matter (storage, contamination, worst-case user safety).

9 Nano-scale public Engram reproduction

Cheng et al. [2026] released only the Engram *architecture* code—no Engram-trained weights are publicly available. To make this paper’s claims reproducible without DeepSeek’s pretraining in-

Table 5: Mini-Engram models released with this paper. All use the same 51.2 M-parameter Engram table (the large preset: $50K \times 256$). Training time on a single NVIDIA RTX PRO 6000 Blackwell, bf16.

	d8 v2	d12 v2	d12@1280	d20@1536
depth $\times$ width	8×512	12×768	12×1280	20×1536
total parameters	178 M	339 M	625 M	1.22 B
scaling-parameters	42 M	110 M	278 M	617 M
ClimbMix tokens	0.50 B	1.32 B	3.34 B	7.40 B
final val bpb	0.912	0.827	0.770	0.730
single-GPU wall-clock	~ 50 min	~ 5 h	~ 10.5 h	~ 70 h

frastructure, we grafted the Engram module into Karpathy’s nanochat training harness [Karpathy, 2026] and trained four Mini-Engram models from scratch on ClimbMix on a single Blackwell GPU (Table 5). To our knowledge these are the first publicly available Engram models. Every measurement in this paper uses the largest of these (d20@1536, 1.22 B parameters) unless explicitly noted.

10 Multi-tenant serving

The layered design is naturally suited to multi-tenant deployment. The shared LoRA is one global artifact loaded once into HBM; the per-user content is a small Engram-row override dictionary that lives in DRAM and is swapped per request via standard tensor indexing (no custom CUDA kernel). On a single Blackwell GPU we measure **232 req/s** on Mini-Engram-d12@1280 (30 users $\times$ 50 facts $\times$ 600 requests), with per-request latency $p50/p99 = 4.2/4.4$ ms and override apply latency $p50 = 0.03$ ms (the override is one embedding, `weight[rows] = ... write`).

Cross-user privacy is zero by construction. User V’s overrides are never in the table when user V is not the active user; a server reads V’s facts from DRAM only on requests routed to V. This is stronger than approaches that require per-user attention masks or per-batch-row routing (e.g. S-LoRA [Sheng et al., 2024]/Punica [Chen et al., 2024]); no custom serving kernel is needed.

A 1 M-user deployment of the layered design needs roughly 100 GB of total storage ($88 \text{ KB} \times 10^6 + 12 \text{ MB}$ shared) and serves at sub-5 ms request latency on commodity hardware; the same population under per-user LoRA is 14.2 TB.

11 Discussion and limitations

The cross-schema result is preliminary. At submission time the cross-schema run is at $n=16/20$ medical patients; the F vs B margin is large enough (33% vs 5% indirect-any, 73% vs 50% indirect-top-1) that the conclusion is unlikely to flip at $n=20$, but the final number should be re-read once the run completes. More importantly, we tested cross-schema generalisation on *one* held-out schema (medical patients) parallel to the personal schema’s indirect-Q types. Whether the shared LoRA generalises to schemas with *different* reasoning structures (e.g. legal compliance, codebases) is open.

Joint training is open. The per-user Engram, per-user LoRA, and shared LoRA in this paper are each trained *independently*. Joint training (per-user Engram co-optimised with the shared LoRA, or the shared LoRA updated with a small fraction of each new user’s content) might give further gains. We chose independent training to keep the substrate decomposition crisp; investigating joint training is the natural next step.

LOCOMO with the layered design is open. Our LOCOMO single-hop results (committed in earlier drafts of this work, on nanochat) use per-user Engram-row insertion alone. Re-running LOCOMO with the layered design (shared LoRA + per-user Engram) is a direct test of whether the meta-skill helps on a published conversational-memory benchmark; we leave this to a follow-up.

Per-user content benchmark is synthetic. Our test users are programmatic, with 30–34 facts each across ~ 8 schemas. The *paraphrase* robustness of Engram-row insertion in the wild (where users

phrase the same fact many different ways at inference time) was characterised in earlier work on the same Engram backbone, but we have not re-tested it with the shared LoRA active.

Capacity of the layered design. We sweep 30–1,000 facts/user in earlier experiments on per-user Engram alone (the per-user joint-OPT density curve drops from 96% top-5 at 100 facts to 72% at 1,000 facts). Whether the shared LoRA changes this curve—either by raising the ceiling or by changing the slope—is not tested in this work.

Choice of shared-LoRA training corpus matters. The 510-sample, 10-user training corpus we used is the smallest plausible size to learn a meta-skill at all. The rank-64 column of Table 1 suggests the corpus may be too small to support a higher-rank LoRA; we expect the rank knob to move with training-corpus scale.

12 Conclusion

Personal memory is two problems, not one. Content is per-user, atomic, and benefits from an edit that is mathematically inert outside its trigger; meta-skill is cross-user, aggregative, and benefits from amortisation across the population. A per-user LoRA conflates them in one substrate—producing a measurable architectural contamination (+1.78 bpb on Mini-Engram-d20 vs the Engram override’s +0.0001 on the same base; the $\sim 15,000\times$ ratio replicates cross-base) and limiting recall to what one low-rank delta can hold.

The fix is to match each sub-problem to its right substrate: **per-user Engram-row override for content** (local by construction, $\Delta\text{bpb} \equiv 0$ on non-trigger inputs) and **one shared LoRA for meta-skill** (cost amortised across users, trained on cross-user “facts-in-context $\rightarrow$ reasoning” samples). On Mini-Engram-d20, this layered architecture Pareto-dominates every per-user single-substrate baseline (100% direct, $7.5\times$ better indirect than per-user LoRA, $4.6\times$ less contamination, 0/20 users worse than the no-adaptor base versus 17/20). It replicates at a smaller Engram, survives a stricter LLM-judge metric, and—preliminarily—generalises across schemas.

The architectural lesson generalises beyond Engram. Any per-user memory substrate that mixes content and meta-skill in one edit incurs the conflation tax we measured; any decomposition that separates them inherits the cleaner Pareto curve. The contribution of this paper is not a new substrate but the right *decomposition* of personal memory into the two substrates we already know how to build well.

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